

A Comprehensive Exploration of Fine-Tuning WavLM for Enhancing Speech Emotion Recognition

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Abstract—Speech Emotion Recognition (SER) is a pivotal area in Human-Computer Interaction (HCI) with numerous applications. Traditional SER models rely on supervised learning but face challenges due to limited labeled data and the subjective nature of emotions. Self-supervised learning (SSL) offers an alternative by leveraging unlabeled audio data. WavLM, a large-scale SSL audio model, has shown promise in various speech processing tasks. This paper investigates the fine-tuning of WavLM for SER, analyzing the impact of different segment of layers on performance. We conducted experiments on the IEMOCAP and RAVDESS datasets, comparing various fine-tuned WavLM models with Wav2Vec 2.0 and HuBERT as SSL baselines. Results reveal that WavLM outperforms SSL baselines, with all-layer fine-tuned WavLM achieving state-of-the-art results. Interestingly, fine-tuning top layers significantly enhances performance, suggesting their role in encoding paralinguistic information. However, fine-tuning all layers remains superior. These findings shed light on optimizing SSL audio models for SER and highlight WavLM's potential in emotion recognition.

Index Terms—Speech emotion recognition, self-supervised learning, audio model, WavLM, fine-tuning

I. INTRODUCTION

Speech emotion recognition (SER) has garnered increasing attention in the realm of Human-Computer Interaction (HCI) in recent years. The ability to detect and interpret emotional cues from human speech is an important aspect in enhancing user experience and making human and machine interactions more natural and robust. SER has diverse applications, ranging from improving call center interactions, mental health assessment to enhancing the effectiveness of spoken dialogue systems.

Commonly, SER models has been approached through supervised learning, where models are trained on labeled emotional speech datasets. In addition, with the help of rapid progress in deep learning, many sophisticated deep learning-based SER models have emerged and displayed favorable outcomes. For instance, convolutional neural networks (CNN) [1, 2], recurrent neural networks [3, 4] and its variation [5, 6]. However, this kind of approach and models often suffers from limitations, including the scarcity of labeled data and the challenges posed by the inherently complex and subjective nature of human emotions. In contrast, self-supervised learning (SSL) has emerged as a promising avenue for training

upstream¹ models in an unsupervised or weakly supervised manner, leveraging the abundance of unlabeled audio data available on the internet to capture underlying patterns and structures of a speech. Subsequently, additional layers can be added on top of pre-trained upstream SSL model to create a downstream² model, which is then trained or fine-tuned using supervised learning on smaller, labeled data for specific tasks.

SSL audio models, such as Wav2Vec 2.0 [7] and HuBERT [8] have shown their effectiveness mainly in automatic speech recognition task (ASR). However, in other tasks, such as speaker verification, speech separation and SER, their performance does not reach the notable levels achieved in ASR. This is because pre-training task of earlier self-supervised learning (SSL) audio models were primarily designed to predict future audio representations based on past representations within an audio sequence, with a focus on capturing phonetic and linguistic content in the audio data. Therefore, these models might not excel in scenarios where the task requires paralinguistic and speaker-specific information. To address this issue, Chen et al. [9] proposed a large scale SSL audio model called WavLM for full-stack speech processing. WavLM applies a dual learning approach during pre-training which not only preserves its capacity to model speech content through masked speech prediction but also enhances its suitability for non-ASR tasks through speech denoising. Furthermore, WavLM scales up the pre-training dataset originally employed for pre-training Wav2Vec 2.0 and Hubert by introducing a broader range of audio backgrounds, an increased number of speakers, and varied speech content.

WavLM attains state-of-the-art outcomes on the SUPERB benchmark [10, 11], a speech processing task benchmark for evaluating SSL audio models, and significantly elevates performance across a range of speech processing tasks, such as speech separation, speaker verification, speaker diarization (task to identify "who spoke when" based on audio segments [12]), and speech emotion recognition (SER), in which the SER results in the benchmark were achieved by freezing transformer layers of WavLM as the upstream model, with

¹"upstream model" typically refers to a model that processes data or executes a particular task before passing its output to another model

²"downstream model" typically refers to a model that is responsible for making predictions or performing a specific task

the actual SER training conducted exclusively on the downstream model. In contrast, certain studies have demonstrated notable enhancements in SER by fine-tuning both the upstream models, such as Wav2Vec 2.0 [13, 14, 15] and HuBERT [16, 17, 18], and the downstream model. However, there has been no prior investigation into exploring WavLM in a similar direction. Fine-tuning WavLM for a particular task, like SER, is anticipated to enable the model's layers to adapt their learned representations to suit the target task. This involves modifying these learned representations at different levels of abstraction, as different layers of WavLM capture diverse information and features relevant to the specific task.

In this paper, we conducted a series of experiments to analyze the performance of fine-tuning various layers of WavLM for speech emotion recognition (SER). Specifically, we employed an all layers fine-tuning and segmented layers fine-tuning. We evaluate the effective of our diverse fine-tuned WavLM SER models on IEMOCAP [19] and RAVDESS [20] datasets, and we compared their performance to fine-tuned versions of Wav2Vec 2.0 and HuBERT, serving as a self-supervised learning (SSL) baseline for SER. Our findings indicate that all layers fine-tuned WavLM achieves state-of-the-art results on both datasets and surpasses Wav2Vec 2.0 and HuBERT as the preferred upstream model for SER. Additionally, our results demonstrate that fine-tuning all layers of the top segment of WavLM outperforms fine-tuning of other segments, indicating that some layers in this segment are particularly instrumental in encoding paralinguistic information within speech.

The rest of this paper is structured as follows: Section 2 introduces the proposed method. Section 3 outlines the details of the experimental setup. Section 4 delves into the presentation of results. Lastly, Section 5 sums up the conclusions drawn from this study and outlines potential future research directions.

II. METHOD

In this paper, we define the Speech Emotion Recognition (SER) task as the mapping of continuous speech data into discrete categorical emotion labels. We first provide an overview of WavLM as the upstream model, then describe of the downstream model's architecture we employ, and conclude with an explanation of the fine-tuning approach for the WavLM layers.

A. WavLM

WavLM [9] is a self-supervised training framework tailored for audio representations, designed to cater to a diverse range of speech processing tasks. Its architecture comprises two primary components: the convolutional encoder layer and the transformer encoder layer [21]. The convolutional encoder consists of seven temporal convolution blocks, each containing 512 channels, followed by GELU normalization and activation layers. This section's role is to encode the audio signal into a sequence of embeddings, utilizing a 20 ms stride and a 25 ms duration receptive field.

The Transformer encoder, includes a convolution-based embedding layer with a kernel size of 128 and encompasses 16 blocks. Notably, this Transformer encoder incorporates the gated relative position bias technique [22], which factors in relative position information within the attention network computation, thereby enhancing the modeling of local audio characteristics. This approach differs from the convolutional relative position embedding technique employed in other self-supervised learning audio frameworks like Wav2Vec 2.0 [22] and HuBERT [8]. The gated relative position bias technique takes into account the content and adaptively adjusts the relative position bias to suit the current speech content being processed. WavLM offers two versions: WavLM Base, featuring 12 transformer encoder layers with 768-dimensional embeddings, and WavLM Large, featuring 24 layers of transformer encoders with 1024-dimensional embeddings. In this paper, we fine-tuned both versions of WavLM.

To enhance the model's performance as a feature extractor across a range of speech processing tasks, WavLM introduces its pre-training framework, which is rooted in noise restoration (denoising) and masked speech prediction techniques, aimed at improving model robustness in challenging acoustic environments and preserving speaker identity quality. During the pre-training phase, WavLM takes as input an audio signal that is intentionally mixed with background noise and/or other speech signals, where approximately 50% of the audio signal is randomly masked. The model's training objective is twofold: first, to discern the primary speaker's voice within the mixed audio, which may include noise or overlapping speech, and second, to predict the information that has been concealed by masking. Drawing inspiration from HuBERT [8], WavLM transforms continuous audio signals into discrete labels, which serve as prediction targets. This transformation is achieved using the K-means method, with the discrete labels being modeled as the prediction targets.

B. Downstream model

In this paper, WavLM acted as the upstream model, in which we add a downstream model on top of it to form an end-to-end SER model. We adopt a simple architecture for our downstream model, which is a mean pooling aggregator along the time direction of the outputs of WavLM's hidden layers, followed by dense layer with the size of neurons that has the same size of WavLM transformer layers' embedding size, in this case 768 for the base version and 1024 for the large version. This layer is then followed by a tanh activation layer, dropout layer with 0.2 probability, and end with a final dense layer supplemented with a softmax activation layer, generating probabilities for each emotion category.

C. WavLM transformer layer fine-tuning strategy

The original WavLM paper reveals that different transformer layers within SSL models hold valuable information for distinct tasks [9]. For instance, the intermediate representations in the top layers of WavLM, which are layers that are closer to downstream model, incorporates linguistic information that are

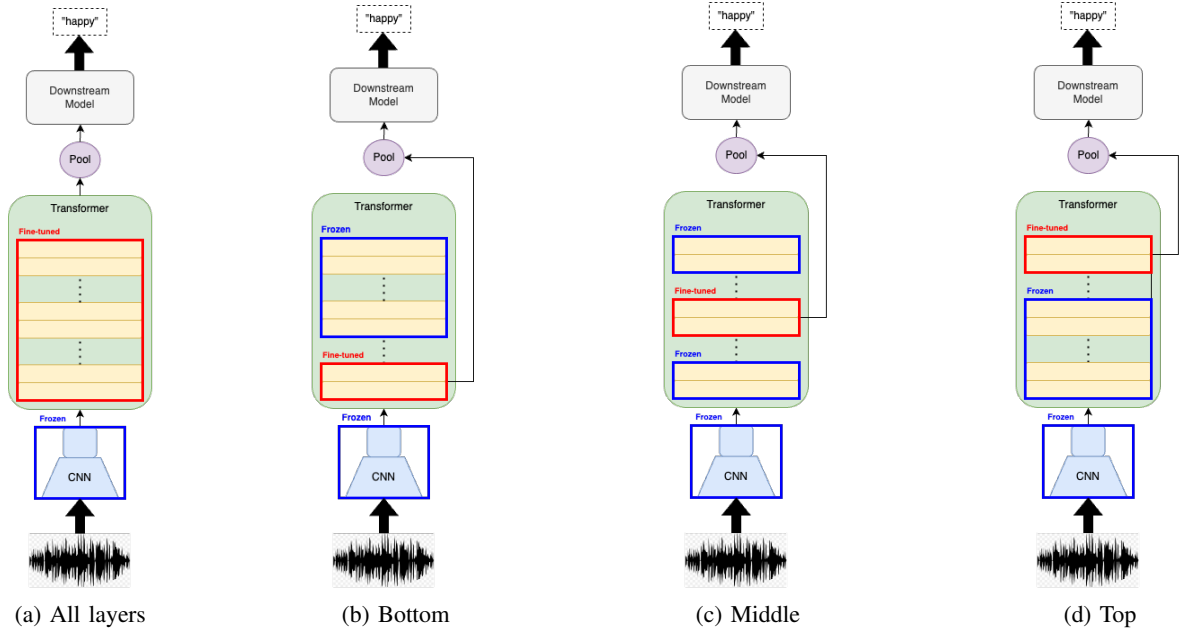


Fig. 1: Segmented fine-tuning and frozen transformer layers configurations. (a): all transformer layers are fine-tuned. (b): the bottom segment layers (several transformer layers near the convolutional layer) are fine-tuned while the middle and top segment are kept frozen. (c): the middle segment layers are fine-tuned while the bottom and top segment are kept frozen. (d): the top segment layers (several transformer layers near the classification head or output layer) are fine-tuned while the bottom and middle segment are kept frozen.

advantageous for speech recognition task, whereas the bottom layers, which are layers that are closer to the input audio, incorporates speaker information that are more effective for tasks like speaker verification. In other research, it has been discovered that middle of Wav2Vec 2.0 transformer layers incorporate paralinguistic information [23], which are useful for tasks like SER.

Inspired with those findings, we aim to investigate the impact of fine-tuning specific transformer layers of the WavLM while keeping others fixed on SER performance. Figure 1 illustrates the various configurations involving trainable and frozen layers that we will explore in this paper. Primarily, we divide the transformer layers of WavLM into three segments: the upper, the middle, and the lower segment. Within each segment, we will designate the layers belonging to that segment as trainable while keeping the layers in the other segments frozen. In all configurations, the convolutional encoder layers will remain in a frozen state. Further details regarding how we group the layers into these three segments will be explained in the subsequent section.

III. EXPERIMENTS

In this section, we outline the experimental setup employed to evaluate the impact of fine-tuning different parts of the WavLM transformer layer on SER performance. We describe the datasets used, training procedure and layer segment grouping.

A. Dataset

1) *IEMOCAP*: The Interactive Emotional Dyadic Motion Capture (IEMOCAP) [19] consist of approximately 12 hours of multimodal data, including speech, text transcriptions, and facial recordings. It consists of 5 sessions involving conversations between 10 different speakers, each session pairing a male actor with a female actress. Annotators assigned emotion labels to each sample, choosing from a predefined emotion set. In order to maintain consistency and facilitate comparison with other studies, we adhere to the IEMOCAP evaluation guidelines outlined in the SUPERB benchmark [10]. This involves combining the "excited" class with the "happy" class, resulting in a dataset comprising 5,531 utterances (consisting of 1,636 happy, 1,103 angry, 1,084 sad, and 1,708 neutral instances), exclusively utilizing the speech data, and evaluating the model's performance through a five-fold cross-validation approach, where each fold corresponds to one session.

2) *RAVDESS*: The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) [20] is a multi-modal dataset featuring 24 actors (12 males and 12 females) delivering two statements, each expressing eight different emotions: happiness, sadness, anger, fear, surprise, disgust, calm, and neutrality. These expressions were both spoken and sung and were repeated twice, resulting in 104 unique vocalizations per actor. To maintain class balance, all eight emotion labels were utilized. Our dataset split, following [24], involved using the first 20 actors for training, actors 20-22 for validation (for early stopping), and actors 22-24 for testing.

we focused exclusively on the speech data from RAVDESS.

B. Layer segment grouping

We employed WavLM Large and Base to evaluate the impact of fine-tuning specific segments within the model. In particular, we divided the total number N transformer layers of WavLM into 3 segments, each containing $N/3$ adjacent layers. Specifically, the bottom segment in WavLM large includes layers 0-7 while in WavLM base includes layer 0-3, the middle segment in WavLM large includes layers 8-15 while in WavLM base includes layers 4-7, and the top segment in WavLM large includes layers 16-23 and in WavLM base includes layers 8-11. For brevity, we denote these fine-tuning configurations as follows:

- **WavLM-large/WavLM-base:** Frozen-layers WavLM large and base, respectively. These are the baselines for comparison our fine-tuned WavLM
- **WavLM-large-A/WavLM-base-A:** All-layers fine-tuned WavLM large and WavLM base, respectively. These are inspired by fine-tuned Wav2Vec 2.0 and HuBERT presented in [16].
- **WavLM-large-B/WavLM-base-B:** Bottom-segment fine-tuned WavLM large and WavLM base, respectively.
- **WavLM-large-M/WavLM-base-M:** Middle-segment fine-tuned WavLM large and WavLM base, respectively.
- **WavLM-large-T/WavLM-base-T:** Top-segment fine-tuned WavLM large and WavLM base, respectively.

Due to computational resource constraints, we restricted fine-tuning of WavLM-large-B, WavLM-large-M and WavLM-large-T to the RAVDESS dataset only and WavLM-base-B, WavLM-large-M and WavLM-base-T to the IEMOCAP dataset only, while WavLM-large-A and WavLM-base-A underwent fine-tuning on both RAVDESS and IEMOCAP.

C. Model checkpoint average

Following [16, 25], in addition to fine-tuning, we employed a technique to reduce output variance in the WavLM models by averaging the top 5 fine-tuned WavLM training checkpoints individually for each of the 5 folds within the IEMOCAP dataset and the RAVDESS train set. The selection of these 5 checkpoints for averaging in each fold was solely based on the WavLM validation-set accuracy, with no access to the test-set during this process.

D. Metrics

For all experiments, we report the Unweighted Accuracy (UA) as the metric. UA calculates the accuracy for each individual class and then takes the average. The formula of UA is as follows:

$$UA = \frac{1}{C} \sum \frac{TP_i}{TP_i + FN_i} \quad (1)$$

where C is the number of emotion classes, TP_i are the true positives or correctly classified instances for class i and FN_i are the false negatives for class i or in other word instances of class i incorrectly classified as other classes.

IV. RESULTS AND DISCUSSIONS

Table 1 provides a summary of the performance of various models in the IEMOCAP dataset for Speech Emotion Recognition (SER). The results are presented as the average UA across all 5 folds. To establish a baseline for comparison, we used the results of Filter Bank (Fbank), Wav2Vec 2.0, and HuBERT, as reported in [16], since they follow the same evaluation method as the SUPERB benchmark. Additionally, we included the UA scores of the completely frozen WavLM base and large models from the SUPERB benchmark leaderboard for SER in the same dataset. Our analysis shows that WavLM-large-A outperforms all the models listed in the table. When compared to other self-supervised learning (SSL) audio models, WavLM exhibits a 0.5% improvement over Wav2Vec 2.0 large and a 1.57% improvement over HuBERT large. Notably, when comparing WavLM-Large-A with the completely frozen WavLM-large and WavLM-Base-A with the completely frozen WavLM-base, we observe a significant 6.74% and 5.82% increase in UA respectively, highlighting the substantial advantage of fine-tuning WavLM for the specific task of SER.

When analyzing the WavLM Base models with fine-tuned segments, it becomes evident that the top segments yield the best performance, outperforming all other segments. When compared to the frozen WavLM-base model, both WavLM-base-T and WavLM-base-M demonstrate superior performance, while WavLM-base-B falls short in terms of performance. This observations suggests that these middle and upper layers play a significant role in learning and fine-tuning intermediate representations crucial for the SER task which requires understanding higher-level emotional and paralinguistic features, while the bottom layers capture low-level, generic features that are more relevant to acoustic and phonetic characteristics. Fine-tuning only the bottom segment may not allow the model to adjust its representations adequately for these specific requirements, while fine-tuning middle and top segments allows the model to specialize its learned features for SER, potentially capturing important emotional cues and nuances. This aligns with findings in [23], which suggest that paralinguistic information in the audio is encoded and learned in the middle-to-upper layers of the model. However, it's worth noting that fine-tuning all layers of WavLM for SER still outperforms fine-tuning only the segments. This indicates that the model takes into account parameters encoded across all layers to effectively recognize emotions from the intermediate representations of the audio.

Table 2 provides an overview of the performance of various models in the RAVDESS dataset for Speech Emotion Recognition (SER). The results are presented as the UA achieved in the test set. To establish a baseline for comparison, we conducted fine-tuning on Wav2Vec 2.0 large and HuBERT large models. The results reveal that WavLM-Large-A also surpasses all the models listed in the table. In comparison to other Self-Supervised Learning (SSL) audio models, WavLM exhibits a notable improvement, with a 1.56% advantage over Wav2Vec